Supplement to "The Cost of Reliable Near-Optimal Set Maintenance with Arriving Arms"

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Notation and scope

The observation model has independent arm streams, each consisting of independent, identically distributed rewards in $[0,1]$, deterministic permanent arrivals $\mathcal{A}_t$, and one observation per physical time step. Each finite $\mathcal{A}_t$ is nonempty, and arbitrarily large finite batches are allowed. Arrivals occur before the action, and reports follow its observation. Let $r^*(t) = \max_{i\in\mathcal{A}_t} r_i$, $\Delta_i(t) = r^*(t) - r_i$, and $\mathcal{G}_t^\varepsilon = \{i \in \mathcal{A}_t : \Delta_i(t) \le \varepsilon\}$. Reports partition $\mathcal{A}_t$ as $(\mathcal{P}_t, \mathcal{N}_t, \mathcal{V}_t)$. Write

$$E_{\text{lab}} = \{\forall t : \mathcal{P}_t \subseteq \mathcal{G}_t^\varepsilon \subseteq \mathcal{A}_t \setminus \mathcal{N}_t\}, \qquad \mathcal{U}_T(\eta) = \sum_{t\le T}\sum_{i\in\mathcal{A}_t} \mathbf{1}\{i \in \mathcal{V}_t, |\Delta_i(t) - \varepsilon| > \eta\},$$

and $R_T = \sum_{t\le T} \Delta_{I_t}(t)$. Reliability means $P(E_{\text{lab}}) \ge 1-\delta$ for every allowed environment. $\mathfrak{R}_\Theta(a)$ minimizes worst-case ER_T over full-model reliable policies with $P_\theta(E_{\text{lab}}, \mathcal{U}_T \le a) \ge 1-\delta$ for all $\theta \in \Theta$. A bar denotes averaging under the independent uniform prior on all candidate signs. These priors are proof devices and impose no Bayesian requirement on the policy.

The probability and sample-budget arguments are given below. Classical change of measure and multiple-recovery scales are credited to the existing literature (Kaufmann et al., 2016; Malloy and Nowak, 2014; Jamieson and Jain, 2018; Mason et al., 2020).

S1 Probability tools and reporting conventions

For the reporting constructions, index arms in arrival order $i = 1, 2, \ldots$, with a fixed order within each batch. Every arm has an independent potential reward stream $(Y_{i,n})_{n\ge1}$ in $[0,1]$. The policy observes its nth entry on its nth pull. Arrival times are deterministic, and reports are issued after the current observation. Write $N_i(t)$ for lifetime observations through round t and $\hat{r}_i(n) = n^{-1}\sum_{s\le n} Y_{i,s}$. The observed reward in physical-time notation is $X_{t,I_t} = Y_{I_t,N_{I_t}(t-1)+1}$. The conclusions below also hold conditionally on an independent exogenous arrival schedule.

For independent $[0,1]$ observations, Hoeffding's inequality gives

$$P\{|\hat{r}_i(n) - r_i| > x\} \le 2e^{-2nx^2}.$$

For completeness, the centered log moment generating function has value and derivative zero at the origin; its second derivative is a variance under an exponentially tilted distribution and is at most $1/4$. Integrating twice gives $Ee^{\lambda(X-EX)} \le e^{\lambda^2/8}$. Multiplication over independent observations and exponential Markov inequality prove the display.

With

$$b_i(n) = \sqrt{\frac{\log(8i^2n^2/\delta)}{2n}},$$

the union of all potential-prefix failures has probability at most $\delta\zeta(2)^2/4 < \delta$. Intersecting the prefix intervals with $[0,1]$ therefore gives simultaneous coverage. On that event each endpoint differs from its mean by at most $2b_i(n)$. No conditioning on a previously selected empirical mean is used. Empty intersections on failure paths are handled by exclusion-first reporting; the reward policy remains defined on all paths.

The checkpoint alternative fixes, before observing arm i,

$$q_i(\rho_i) = \left\lceil \frac{2}{\rho_i^2} \log \frac{4i(i+1)}{\delta} \right\rceil.$$

Before q_i observations its interval is $[0,1]$; afterwards it is permanently

$$[\hat{r}_i(q_i) - \rho_i/2, \hat{r}_i(q_i) + \rho_i/2] \cap [0,1].$$

The sum of coverage failures is at most $\sum_i \delta/\{2i(i+1)\} = \delta/2$. On coverage, each endpoint differs from the true mean by at most ρ_i. These intervals are nonempty on every path and only contract once. If $\underline{r}^*(t) = \max_i \ell_i(t)$ and $\overline{r}^*(t) = \max_i u_i(t)$, use

$$\mathcal{P}_t = \{i : \ell_i(t) \ge \overline{r}^*(t) - \varepsilon\}, \quad \mathcal{N}_t = \{i : u_i(t) < \underline{r}^*(t) - \varepsilon\}, \quad \mathcal{V}_t = \mathcal{A}_t \setminus (\mathcal{P}_t \cup \mathcal{N}_t).$$

The sets $\mathcal{P}_t, \mathcal{N}_t$ are disjoint because $\underline{r}^*(t) \le \overline{r}^*(t)$ and $\ell_i \le u_i$. Coverage implies $\mathcal{P}_t \subseteq \mathcal{G}_t^\varepsilon \subseteq \mathcal{A}_t \setminus \mathcal{N}_t$. Moreover, $\mathcal{N}_t$ is increasing as observations and arrivals accumulate: $\underline{r}^*(t)$ cannot decrease and an existing u_i cannot increase.

For two Bernoulli mean vectors $\boldsymbol{r} = (r_i)_i$ and $\boldsymbol{r}' = (r_i')_i$ and a bounded stopping time τ, action kernels, policy randomization, and stopping kernels cancel in the history likelihood ratio. Padding after stopping by null symbols and taking iterated expectations yields

$$D(P_{\boldsymbol{r}}^{H_\tau} \| P_{\boldsymbol{r}'}^{H_\tau}) = \sum_i E_{\boldsymbol{r}} N_i(\tau)\, \mathrm{kl}(r_i, r_i'). \tag{1}$$

The same proof applies to arbitrary observation laws with finite two-way KL. For an event F measurable at stopping, splitting the likelihood integral over F, F^c and applying Jensen gives

$$D(P\|Q) \ge \mathrm{kl}(P(F), Q(F)) \ge P(F) \log \frac{1}{Q(F)} - \log 2.$$

The second inequality uses the binary entropy bound $h(p) \le \log 2$. We repeatedly use

$$2(p-q)^2 \le \mathrm{kl}(p,q) \le \frac{(p-q)^2}{q(1-q)}.$$

The upper bound follows from $\log x \le x-1$; for the lower bound the second derivative in p is $1/\{p(1-p)\} \ge 4$ and both value and derivative vanish at $p=q$.

S2 Joint guarantee of Checkpoint-UCB

Checkpoint-UCB fixes each arm's checkpoint before seeing its data. At even physical rounds it samples the earliest-arriving arm below its lifetime checkpoint, if one exists. At all other rounds it maximizes, over every available arm including reported exclusions,

$$J_i(t) = \hat{r}_i(N_i(t-1)) + \sqrt{3\log(2t)/N_i(t-1)},$$

with index $+\infty$ at zero count and deterministic ties. Every observation serves both the lifetime checkpoint and the empirical reward estimate. The fixed-checkpoint argument in Section S1 gives $P(E_{\mathrm{lab}}) \ge 1-\delta/2$. If $\mathcal{I}_T$ denotes designated service rounds, then $|\mathcal{I}_T| \le \sum_{i \le K_T} q_i$, because each such observation is charged to an arm below its single lifetime quota. The following reward argument does not assume separated arrival windows.

A reward bound that does not condition on report coverage. At physical time t, order arms by their first observed rewards. At most t streams have been exposed, each at at most t potential prefixes. A previously unobserved stream is independent of the history selecting it. Applying Hoeffding to each newly exposed stream and union bounding gives

$$P\left\{\exists \text{ observed arm and prefix } n \le t : |\hat{r}_i(n) - r_i| > \sqrt{3\log(2t)/n}\right\} \le 2t^2(2t)^{-6} = \frac{1}{32t^4}.$$

The exposure order is used only for this union bound, not to assert independence among observed counts. An ordinary UCB pull with finite index, on this event's complement, has $\Delta_{I_t}(t) \le 2\sqrt{3\log(2t)/N_{I_t}(t-1)}$. Each arm–pre-pull-count pair is charged at most once; interleaved service observations skip counts. Therefore the ordinary, nonfirst-pull regret is at most

$$4\sqrt{3\log(2T)}\sum_i \sqrt{N_i(T)} \le 4\sqrt{3K_T T\log(2T)}$$

on good rounds. First pulls cost at most K_T, and the sum of expected bad-round losses is less than one. Adding the actual regret of service rounds proves

$$ER_T \le \min\left\{T, K_T + 4\sqrt{3K_T T\log(2T)} + 1 + E\sum_{t\in\mathcal{I}_T}\Delta_{I_t}(t)\right\}.$$

There is no δT term: the reward policy never permanently discards an arm because of a reporting error.

Lemma S2.1 (Completion under arbitrary arrivals)**.** *Fix $0 < \eta < 1$ and use Checkpoint-UCB with the predetermined precision $\rho = \eta/4$. Let $\overline{Q}_t$ be the post-service workload of the virtual FIFO queue that inserts q_i units at arm i's arrival and serves one unit at every even physical round when nonempty. On the simultaneous checkpoint-coverage event, for each fixed deterministic endpoint T,*

$$\mathcal{U}_T(\eta) \le \sum_{t \le T} K_t 1\{\overline{Q}_t > 0\} \le \min\{K_T T, 2K_T \sum_{i \le K_T} q_i\}.$$

Proof. If all current checkpoints are complete, all endpoint errors are at most $\eta/4$ and every arm whose boundary distance exceeds η has a determinate report. A virtual FIFO queue adding q_i work at each arrival and serving one unit at every even round dominates unfinished checkpoints; additional UCB observations only reduce actual work. Its busy periods, including alignment to the first even round, have total length at most $2\sum_i q_i$. Thus on simultaneous coverage the unresolved area is bounded by the queue-weighted sum in the statement, which is at most $K_T T$ and at most $2K_T \sum_{i \le K_T} q_i$. □

S3 Broadcast uncertainty and queue area

Precise claim. Let $0 < \varepsilon \le 1/16$, $0 < \delta \le 1/16$, $0 < \eta < \varepsilon/2$. The reference arm has mean $1/4 + \varepsilon/2$. All candidates have mean $p = 1/4$ in environment zero; environment j raises only candidate j to $p + 2\varepsilon$. Let M_t be the candidate count, a_t the new-candidate count, and

$$q_- = \left\lfloor \frac{\mathrm{kl}(1/4, \delta)}{\mathrm{kl}(1/4, 1/4 + 2\varepsilon)} \right\rfloor, \quad Q_t = (Q_{t-1} + q_- a_t - 1)_+, \quad Q_0 = 0,$$

$$\mathcal{B}_T = \sum_{t \le T} (M_t - 1)_+ 1\{Q_t > 0\}.$$

Every policy reliable on these environments obeys

$$E_0 \mathcal{U}_T(\eta) \ge (3/4 - \delta)\mathcal{B}_T, \qquad P_0\{\mathcal{U}_T(\eta) \ge \mathcal{B}_T/4\} \ge 1/4$$

when $\mathcal{B}_T > 0$.

Proof. The denominator defining q_- is at most $\mathrm{kl}(1/4, 3/8) \le 1/15$, whereas its numerator is at least $\frac{1}{4}\log 4 + \frac{3}{4}\log(4/5) \ge 7/48 > 2/15$. Thus $q_- \ge 2$. The recursion gives, by induction,

$$Q_t = \max_{0 \le u \le t} \left\{ q_- \sum_{r=u+1}^{t} a_r - (t - u) \right\}.$$

If $Q_t > 0$, there is a suffix in which total candidate demand exceeds all physical observation slots. Its arms had no observations before that suffix. Consequently at least one deterministic candidate

label j satisfies $E_0 N_j(t) < q_-$. The choice is made from expected counts, not from a realized history. For any other candidate i, membership of i in $\mathcal{P}_t$ is erroneous under environment j. Its probability there is at most δ. The history KL is strictly below $\mathrm{kl}(1/4, \delta)$, so data processing and monotonicity of binary KL force $P_0(i \in \mathcal{P}_t) < 1/4$. Since all candidates are genuinely good under environment zero, $P_0(i \in \mathcal{N}_t) \le \delta$. It follows that $P_0(i \in \mathcal{V}_t) \ge 3/4 - \delta$. These candidates all have boundary distance $\varepsilon/2 > \eta$. Summing at deterministic busy rounds proves the expectation bound, entirely under the one environment zero.

Let Y count candidate unresolved area only at busy rounds with $M_t \ge 2$. Then $Y \le \mathcal{U}_T$, $0 \le Y \le 2\mathcal{B}_T$, and $E_0 Y \ge 11\mathcal{B}_T/16$. If $p = P_0(Y \ge \mathcal{B}_T/4)$, then $11\mathcal{B}_T/16 \le \mathcal{B}_T/4 + (7\mathcal{B}_T/4)p$, so $p \ge 1/4$. □

Queue area. For an integer $q \ge 2$, suppose m candidates arrive by time s and put $D = s + qm - 1$. Then

$$\sum_{t \le D} (M_t - 1)_+ 1\{Q_t(q) > 0\} \ge \frac{q-1}{2} m(m-1).$$

To prove this, remove all subsequent arrivals, which can only decrease both the queue and its weights. The qm service units finish by D. At the nth actual service, at least $\lceil n/q \rceil$ candidates have arrived, so service rounds weighted by $M_t - 1$ sum to at least $\sum_{n=1}^{qm} (\lceil n/q \rceil - 1) = qm(m-1)/2$. The post-service queue is zero at the last service round of each busy period. Their cumulative arrival counts are distinct integers in $\{1, \dots, m\}$, so removing their weights costs at most $m(m-1)/2$. This proves the claim.

In particular, if along a fixed infinite schedule a policy reliable in every environment has a deterministic area envelope $a_T = o(T)$, then $K_t = o(\sqrt{t})$. Otherwise along s_n there are at least $c\sqrt{s_n}$ candidates. Choose the first $m_n = \lfloor c\sqrt{s_n} \rfloor$; eventually $q_- m_n \le s_n$. The queue bound and the preceding probability statement at $2s_n$ give $P_0(\mathcal{U}_{2s_n} \ge c' s_n) \ge 1/4$, contradicting $\delta \le 1/16$ and the proposed envelope.

S4 Partial simultaneous recovery

Precise claim. Let $m \ge 1$ independent uniform signs specify independent observation laws P_i^+, P_i^-. All nuisance parameters and observations are independent of these signs. Sampling may be adaptive and interleaved, and τ is bounded. At every sign configuration, the probability of any erroneous determinate report is at most $\delta \le 1/16$. Let J_τ be the number of determinate reports and $\kappa = \max_i \max\{D(P_i^+ \| P_i^-), D(P_i^- \| P_i^+)\} \in (0, \infty)$. If $\overline{P}(J_\tau \ge m/2) \ge 1/2$, then

$$\overline{E} \sum_i N_i(\tau) \ge \frac{m}{1024\kappa} \log \frac{m}{\delta}.$$

Proof. Let L_i be the observed positive-versus-negative log likelihood ratio for coordinate i. After conditioning on the full stopped history, the action and stopping kernels cancel. The likelihood factors over coordinates and so does the posterior, even though the sample counts need not be independent. The smallest coordinatewise posterior error is $e_i = (1 + e^{|L_i|})^{-1}$. If S is the reported

subset, its conditional probability of complete correctness is at most $\prod_{i\in S}(1-e_i)$; choosing a non-MAP label cannot improve it. Thus

$$\overline{E}\left[1-\prod_{i\in S}(1-e_i)\right]\le\delta.$$

With probability at least $1-2\delta$ this product is at least $1/2$ and hence $\sum_{i\in S}e_i\le\log 2$. Intersecting with $J_\tau\ge m/2$ costs at most 2δ and leaves probability at least $3/8$. For $m\ge 4096$, Jensen gives on this intersection

$$\sum_i|L_i|\ge\sum_{i\in S}\{\log(1/e_i)-\log 2\}\ge\frac{m}{2}\log\frac{m}{4\log 2}.$$

Successive conditional expectations of signed likelihood increments give

$$\overline{E}\sum_i\sigma_iL_i\le\kappa\overline{E}\sum_i N_i(\tau).$$

Finite two-way KL makes the stopped increments integrable. Given the history, $E(\sigma_iL_i\mid H_\tau)=|L_i|(1-2e_i)$, and $2|L_i|e_i\le 1$ for every real L_i. Therefore

$$\kappa\overline{E}\sum_i N_i\ge\overline{E}\sum_i|L_i|-m\ge m\left\{\frac{3}{16}\log\frac{m}{4\log 2}-1\right\}\ge\frac{m}{256}\log m,$$

where the last inequality holds for $m\ge 4096$.

For the confidence term, fix a sign vector and flip coordinate i only. The event reporting the original sign on coordinate i has original probability at least its determinate-report probability minus δ, and flipped probability at most δ. Average the data-processing inequalities over vectors and coordinates and use joint convexity of binary KL. Since $\overline{E}J_\tau\ge m/4$, this yields

$$\kappa\overline{E}\sum_i N_i\ge m\,\mathrm{kl}(1/4-\delta,\delta)\ge\frac{m}{256}\log(1/\delta).$$

To check the last scalar inequality, write $L=\log(1/\delta)$. For $L\le 8$, Pinsker gives a lower bound $2(1/4-2\delta)^2\ge 1/32\ge L/256$. For $L\ge 8$, binary entropy at most $\log 2$ gives $\mathrm{kl}(1/4-\delta,\delta)\ge 3L/16-\log 2\ge L/16$. When $m\ge 4096$, the maximum of the two preceding bounds is at least half their sum. When $m<4096$, $\log m\le 3\log(1/\delta)$, and the confidence bound alone suffices. Both cases imply the stated constant $1/1024$. □

Single-window truncation. If such a batch has H rounds with qualifying boundary distance larger than η, set

$$h_*=\min\left\{H,\left\lceil\frac{m\log(m/\delta)}{1024\kappa}\right\rceil-1\right\}.$$

At every $r\le h_*$ there are at most r batch observations. The recovery lemma's contrapositive makes the probability of half-completion less than $1/2$, so the expected unresolved count is at least $m/4$. Thus batch area has expectation at least $mh_*/4$. Its first-h_*-round area Y lies in $[0,mh_*]$;

consequently $\overline{P}(Y \geq mh_*/8) \geq 1/7$. One fixed sign configuration attains at least this probability. If $h_* = 0$, only the trivial zero lower bound is asserted.

S5 Worst-schedule area and growth consequences

Suppose $0 < \varepsilon \leq 1/8$, $0 < \eta \leq \varepsilon/32$, $0 < \delta \leq 1/16$, and $K \geq 2$. Let $\mathfrak{A}_{T,K}$ be the smallest area threshold attainable with joint label-and-area probability at least $1 - \delta$, uniformly over reward environments and schedules with $K_T \leq K$. The policy must be reliable on the full model.

Upper bound. Use fixed precision $\rho = \eta/4$. Then

$$q_i = \left\lceil 32\eta^{-2} \log \frac{4i(i+1)}{\delta} \right\rceil \leq 99\eta^{-2} \log(K/\delta), \qquad i \leq K.$$

Lemma S2.1 proves, on coverage,

$$\mathcal{U}_T \leq \min\{KT, 2K \sum_{i \leq K} q_i\} \leq 256 \min\{KT, K^2\eta^{-2} \log(K/\delta)\}.$$

The unconditional reward bound in Section S2 and at most $\sum_i q_i$ service pulls also give

$$ER_T \leq \min\{T, K + 4\sqrt{3KT \log(2T)} + 1 + 99K\eta^{-2} \log(K/\delta)\}.$$

Lower bound. At round one release a reference of mean $1/2 + \varepsilon$ and $m = K - 1$ candidates of means $1/2 \pm 2\eta$. Their signs are independent and specify their relative-goodness labels. The reference is always optimal; its known distribution gives no sign information. Boundary distances equal $2\eta > \eta$, and two-way KL is at most $128\eta^2$. Set

$$q_* = \frac{m \log(m/\delta)}{131072\eta^2}, \qquad h = \min\{T, \lceil q_* \rceil - 1\}.$$

Since $\eta \leq 1/256$, $q_* \geq m \log(m/\delta)/2 > 1$ and $h \geq \frac{1}{2} \min\{T, q_*\}$. Single-window truncation gives a fixed environment with

$$P\{\mathcal{U}_T \geq mh/8\} \geq 1/7 > \delta.$$

Therefore no valid uniform area quantile can be smaller than $m \min\{T, q_*\}/16$. The comparisons $m \geq K/2$ and $\log(m/\delta) \geq \frac{1}{2} \log(K/\delta)$ yield

$$\mathfrak{A}_{T,K} \geq 2^{-24} \min\{KT, K^2\eta^{-2} \log(K/\delta)\}.$$

The lower bound uses a static first-round burst; it does not attribute its statistical logarithm uniquely to dynamic arrivals.

At fixed task parameters, uniformly over schedules satisfying a terminal count limit $K_T \leq k(T)$, sublinear expected regret and sublinear area envelopes are simultaneously attainable if and only

if $k(T)^2 \log k(T) = o(T)$. Necessity is the last lower bound; sufficiency follows from the displayed upper bounds. This terminal-count-class conclusion allows the worst schedule to depend on T.

Growth on a fixed schedule. At fixed η, δ, the checkpoints are $O(\log(ei))$, so $K_T^2 \log(eK_T) = o(T)$ suffices for both sublinear area and expected regret. Combined with the broadcast consequence in Section S3, a path with $K_t \asymp t^\alpha$, $0 < \alpha < 1$, admits both if and only if $\alpha < 1/2$. This is an exponent statement and does not remove the critical logarithm.

A fixed sparse schedule that still defeats a uniform area envelope. Set

$$s_n = 2^{2^n}, \qquad m_n = \left\lfloor \frac{\sqrt{s_n}}{(\log s_n)^{1/4}} \right\rfloor.$$

Release one reference at time one and m_n candidates at each s_n. Earlier batch sizes sum to $o(m_n)$, so $K_{s_n}/\sqrt{s_n} \to 0$; between batches the ratio decreases. Thus $K_t = o(\sqrt{t})$ on this one schedule. For each selected batch n, give the reference mean $1/2 + \varepsilon$, that batch means $1/2 \pm 2\eta$, and other candidates mean $1/2$. Let

$$h_n = \left\lceil \frac{m_n \log(m_n/\delta)}{131072\eta^2} \right\rceil - 1, \qquad T_n = s_n + h_n - 1.$$

For large n, $h_n = o(s_n)$ and the next batch has not arrived. The old history is independent of the new signs. Single-window truncation supplies one fixed configuration for this endpoint with $P(\mathcal{U}_{T_n} \geq m_n h_n/8) \geq 1/7$. But

$$\frac{m_n h_n}{T_n} \asymp_{\eta,\delta} \sqrt{\log s_n} \longrightarrow \infty.$$

No deterministic envelope $a_T = o(T)$ can consequently hold uniformly over reward environments for this schedule. The reward environment may vary with the chosen endpoint; this is exactly the uniform-in-environment statement and is not a claim of simultaneous failure at all endpoints in one fixed reward environment.

S6 Area-constrained temporary tasks

In window e, suppose M_e candidates have gaps $\varepsilon - \sigma_{e,k} d_e$, and a new reference makes their signs irrelevant after that window. Put $w_e = M_e H_e$, $V = \sum_e w_e$, and let p_e be the product-prior probability that at least half of batch e becomes determinate simultaneously before its window ends. Define

$$\Phi_\delta(u) = \begin{cases} \mathrm{kl}(1 - \delta - u, \delta), & u < 1 - 2\delta, \\ 0, & u \geq 1 - 2\delta, \end{cases}$$

$$g_{m,\delta}(p) = \max\left\{ 0, \frac{(p - 2\delta)_+}{2} \left[\log \frac{m}{4 \log 2} \right]_+ - 1, \Phi_\delta(1 - p/2) \right\}.$$

Each displayed component is convex and nondecreasing in p.

Proof of the general area-to-regret lower bound. If window e never half-completes, at every round more than half its candidates remain unresolved. Its area is at least $w_e/2$. On area success, $\sum_e w_e 1_{S_e^c} \le 2a$; outside that event this sum is at most V. Consequently

$$\sum_e w_e p_e \ge (1-\delta)V - 2a.$$

Stop batch e at first half-completion or at its endpoint. The likelihood argument above now intersects the product-posterior event with completion probability p_e, giving intersection probability at least $(p_e - 2\delta)_+$ and expected determinate count at least $M_e p_e/2$. The absolute-likelihood and coordinate-flip derivations therefore give

$$\kappa_e \overline{E} N_e \ge M_e g_{M_e,\delta}(p_e).$$

For the absolute-likelihood term, when the bracketed logarithm is positive its Jensen expression is increasing for subset sizes at least $M_e/2$; when it is nonpositive the stated term is at most zero. Each observation counted before stopping has regret at least $\varepsilon - d_e$. The signs of all batches belong to one product prior, and samples of different batches are disjoint. Hence

$$\sup_\sigma E_\sigma R_T \ge \inf_{\substack{0\le p_e\le 1 \\ \sum_e w_e p_e \ge (1-\delta)V - 2a}} \sum_e \frac{(\varepsilon - d_e)M_e}{\kappa_e} g_{M_e,\delta}(p_e). \tag{2}$$

This is an information lower relaxation, not a claim that every feasible allocation is achieved by a causal policy. □

Homogeneous consequence. Take $M_e = M$, $H_e = H$, $d_e = d$, and $a = JMH/8$. On success at least $3J/4$ windows half-complete; thus $\sum_e p_e \ge 45J/64$. At least $3J/8$ windows have $p_e \ge 1/2$: otherwise the sum is at most $3J/8 + (5J/8)/2 = 11J/16 < 45J/64$. The Bernoulli means lie in $[1/4, 3/4]$, giving $\kappa_e \le 32d^2$. The partial-recovery lemma and $\varepsilon - d \ge \varepsilon/2$ yield

$$\sup_\sigma E_\sigma R_T \ge \frac{3J}{8} \frac{\varepsilon M}{2 \cdot 1024 \cdot 32 d^2} \log \frac{M}{\delta} \ge 2^{-20} \frac{\varepsilon J M}{d^2} \log \frac{M}{\delta}.$$

S7 Shared evidence and rapid/deferred endpoints

S7.1 Shared evidence across separated windows

Let batch e have reference mean b_e, M_e candidate means $b_e - \varepsilon \pm d_e$, and common predetermined precision ρ_e. The final batch contains only its reference; set $M_J = M_{-1} = 0$ and $S_e = \sum_{i \in F_e} q_i(\rho_e)$. Assume

$$H_e \ge 2S_e, \quad b_J - b_0 \le \varepsilon/4, \quad d_e + \rho_e + \rho_{e+1} < b_{e+1} - b_e, \quad 2\max_e \rho_e \le \varepsilon/2.$$

The algorithm is not told which arm is the reference or when the next window starts. By induction every batch completes its checkpoints within its first $2S_e$ rounds, before the next batch appears. On coverage, successive reference levels satisfy $b_e + \rho_e < b_{e+1} - \rho_{e+1}$. Once batch e completes, the maximum upper endpoint of all available arms is at most $b_e + \rho_e$. Its candidates are therefore completely classified when $2\rho_e < d_e$.

An old candidate in batch $e-1$ has upper endpoint at most $b_{e-1} - \varepsilon + d_{e-1} + \rho_{e-1}$, whereas the new reference has lower endpoint at least $b_e - \rho_e$. Their difference exceeds ε, excluding the entire preceding candidate batch without any further old-arm observations. Older exclusions persist. Every reference has gap at most $\varepsilon/4$ and total endpoint error at most $\varepsilon/2$, so all references become positively certified. If $F = \{e < J : 2\rho_e < d_e\}$, the area is consequently at most

$$\sum_{e \notin F} M_e H_e + 2 \sum_{e \in F} M_e S_e + 2 \sum_{e=0}^{J} (e + 1 + M_{e-1}) S_e.$$

The terms cover current nonfine candidates, fine candidates, and temporarily unresolved references plus the preceding candidate batch. All earlier candidate batches have already been excluded.

For expected regret, every observation with pre-pull count below q_i occurs in its arm's birth window. Such reference observations have zero regret, and candidate observations cost at most $\varepsilon + d_e$ each. All observations with pre-pull count at least q_i are ordinary UCB pulls. For each arm that has a positive gap by T, let $\gamma_i = \min_{t \le T : i \in \mathcal{A}_t,\, \Delta_i(t) > 0} \Delta_i(t)$ and $D_i^T = \max_{t \le T : i \in \mathcal{A}_t} \Delta_i(t)$. Use the good-round concentration bound from Section S2. At a good UCB round, any positive-gap pull has pre-pull count at most $12 \log(2T)/\gamma_i^2$. Since lifetime counts do not repeat, this gives the unconditional bound

$$ER_T \le \sum_{e<J} \sum_{k=1}^{M_e} (\varepsilon + d_e) q_{e,k} + \sum_{i : \gamma_i \text{ exists}} D_i^T \left[\left\lceil \frac{12 \log(2T)}{\gamma_i^2} \right\rceil - q_i + 1 \right]_+ + 1. \tag{3}$$

The completion-before-next-batch assumption is essential to the first charging term. It is not an assertion about arbitrary overlapping arrivals.

S7.2 Rapid and deferred endpoints on one family

Here $J \ge 1$, $M \ge \max\{2, J\}$, $0 < \varepsilon \le 1/8$, $0 < \delta \le 1/16$, $Jv \le \varepsilon/4$, $0 < d \le v/32$, and $\eta = d/2$. The reference means are $b_e = 1/2 + ev$ and the M candidates born in window $e < J$ have means $b_e - \varepsilon + \sigma_{e,k} d$. Each window has length H, and the final window contains only a new reference. Set

$$K = J(M+1) + 1, \quad L_K = \log \frac{4K(K+1)}{\delta}, \quad q_f = \lceil 32 L_K / d^2 \rceil, \quad q_c = \lceil 128 L_K / v^2 \rceil,$$

$$H = 128(M + J + 1) q_f, \qquad T = (J+1)H, \qquad V = JMH.$$

These are analysis quantities: the policy uses the selected precision $\rho_f = d/4$ or $\rho_c = v/8$, not K, H, T or the best-arm identities. Both choices satisfy the separated-window conditions.

The preceding area bound is at most $B_{J,M} q_f$ for the fine policy and $V + B_{J,M} q_c$ for the coarse

policy, where

$$B_{J,M} = 2(M+1)\{J(2M+1) + J(J-1)/2\} + 2(M+J+1).$$

For $2 \le M$ and $J \le M$, $B_{J,M} \le 12JM^2$. Indeed, $M+1 \le 3M/2$, $2M+1 \le 5M/2$, and $J-1 \le M$ bound the main term by $9JM^2$, and the final term by $3JM^2$. The definition of H gives $B_{J,M} q_f \le V/8$, while $q_c \le q_f$ gives coarse area at most $9V/8$. Both statements hold on the same fixed-checkpoint coverage event of probability at least $1 - \delta/2$.

For rapid maintenance the area-to-regret lower bound applies. For its upper bound, every available arm in this family has gap at most 2ε. Hence, on every path, $R_T \le 2\varepsilon T$. Since $(J+1)(M+J+1)/(JM) \le 5$,

$$2\varepsilon T \le 1280 \varepsilon JM q_f.$$

Furthermore $K \le 2M^2$, $K+1 \le 3M^2$ imply $L_K \le 5\log(M/\delta)$, and $q_f \le 165 d^{-2} \log(M/\delta)$. Thus

$$2^{-20} \frac{\varepsilon JM}{d^2} \log \frac{M}{\delta} \le \mathfrak{R}_\Theta(V/8) \le 2^{18} \frac{\varepsilon JM}{d^2} \log \frac{M}{\delta}.$$

This upper bound uses the stated horizon and must not be extended unchanged to arbitrarily long windows.

For the coarse policy, within-checkpoint candidate regret is at most $2\varepsilon JM q_c$. Candidates have $\gamma_i \ge \varepsilon - d \ge 3\varepsilon/4$, $D_i^T \le 2\varepsilon$; old references have $\gamma_i \ge v$, $D_i^T \le Jv$. Dropping the favorable quota subtraction in (3) and absorbing integer terms gives

$$\sup_\Theta E R_T \le 2\varepsilon JM q_c + 64 \left(\frac{JM}{\varepsilon} + \frac{J^2}{v} \right) \log(2T) + 1.$$

At fixed $J, M, \varepsilon, v, \delta$, $d \downarrow 0$ makes $T = \Theta(d^{-2})$. Consequently the rapid endpoint is $\Theta(d^{-2})$ and the deferred endpoint is at most $O(\log(1/d))$. No matching lower bound for the latter endpoint is asserted.

S8 Regret quantiles under the same maintenance protocol

The following result strengthens the expectation lower bound. Its statement uses the same reporting event and the same cumulative unresolved area; it does not introduce a separate deadline requirement. Write $\overline{P}$ for the product prior on all candidate signs in the repeated-update family.

Lemma S8.1 (Completion with a bounded candidate-sampling budget)**.** *Consider one batch of m candidates in its validity window, under the assumptions of the partial-recovery lemma. Let τ be the first round at which at least $m/2$ candidates have determinate reports, or the end of the window if this does not occur. Let S denote that completion occurs, and let $N(\tau)$ count only samples of this batch. Put*

$$L = \frac{m \log(m/\delta)}{1024\kappa}, \qquad q = \lfloor L/2 \rfloor.$$

If $L \geq 4$, then

$$\overline{P}\{S,\ N(\tau) \leq q\} < \frac{1}{2}.$$

Proof. Stop at the earliest of τ, the window endpoint, and the first round with $q+1$ candidate samples. At this bounded stopping time ζ, the number of candidate samples is at most $q+1 < L$. On $\{S, N(\tau) \leq q\}$, the budget stopping condition has not fired before τ, so $\zeta = \tau$ and at least $m/2$ reports are determinate. If the displayed event had probability at least $1/2$, the partial-recovery lemma at ζ would give $\overline{E}N(\zeta) \geq L$, contradicting the deterministic bound $N(\zeta) \leq q+1 < L$.

The extra sample in the stopping rule is needed. Stopping exactly at the qth candidate sample would miss a completion that uses q candidate samples and later observations of the reference arm. □

Theorem S8.2 (Regret quantiles required by rapid maintenance)**.** *In the repeated-update family with J validity windows, M candidates per window, $M \geq \max\{2, J\}$, $Jv \leq \varepsilon/4$, $d \leq v/32$, $\eta = d/2$, and $0 < \delta \leq 1/16$, suppose a policy is δ-reliable on the family and, for every sign configuration σ,*

$$P_\sigma\{E_{\mathrm{lab}},\ \mathcal{U}_T(\eta) \leq JMH/8\} \geq 1 - \delta.$$

Then there is one fixed configuration σ for which

$$P_\sigma\left\{R_T \geq 2^{-20}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}\right\} \geq \frac{11}{80}.$$

This conclusion permits the policy to know the reference identities, the arrival schedule, and the horizon.

Proof. Let S_e denote completion of at least half of the current candidates before the next reference arrives, and let N_e be that batch's sample count at its first such completion, or at the window endpoint when completion fails. Set

$$L_0 = \frac{M \log(M/\delta)}{32768 d^2}, \qquad q_0 = \lfloor L_0/2 \rfloor.$$

The two directional Bernoulli divergences are at most $32d^2$, so the partial-recovery lower bound is at least L_0. The assumptions imply $d \leq 1/1024$ and $M \geq 2$, hence $L_0 > 4$. The preceding budget argument, with $q_0 + 1 < L_0$, gives

$$\overline{P}(F_e) < \frac{1}{2}, \qquad F_e := \{S_e, N_e \leq q_0\}.$$

The argument is valid with all previous-window information included in the history. Those observations and later reference distributions carry no information about this batch's independent signs.

On the area-success event, any window without S_e contributes at least $MH/2$ area. Consequently $\sum_e 1_{S_e} \geq 3J/4$ on an event of prior probability at least $1-\delta$. If $F = \sum_e 1_{F_e}$, then $\overline{E}F \leq J/2$ and Markov's inequality gives

$$\overline{P}\{F \leq 5J/8\} \geq 1/5.$$

No independence across windows is asserted or needed. Intersecting these two events has probability at least $1/5 - \delta \geq 11/80$. On their intersection at least $J/8$ windows complete with $N_e > q_0$, and therefore with $N_e > L_0/2$. Each candidate observation in its birth window has gap at least $\varepsilon - d \geq \varepsilon/2$. These are observations of distinct batches in disjoint windows. Pathwise on that intersection,

$$R_T \geq \frac{\varepsilon}{2}\frac{J}{8}\frac{L_0}{2} = 2^{-20}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}.$$

A mixture probability at least $11/80$ implies that one fixed configuration has at least that probability. Thus no different worst environment is selected for different windows. □

Corollary S8.3 (Rapid maintenance with a joint regret quantile)**.** *Let $\mathfrak{R}_\Theta^{\mathrm{joint}}(a)$ be the infimum of r over policies that are reliable on the full bounded model and satisfy*

$$\inf_{\theta\in\Theta} P_\theta\{E_{\mathrm{lab}}, \mathcal{U}_T(\eta) \leq a, R_T \leq r\} \geq 1-\delta.$$

For the repeated-update horizon $H = 128(M+J+1)\lceil 32d^{-2}\log\{4K(K+1)/\delta\}\rceil$, where $K = J(M+1)+1$, universal positive constants c, C satisfy

$$c\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta} \leq \mathfrak{R}_\Theta^{\mathrm{joint}}(JMH/8) \leq C\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}.$$

Proof. For the lower bound, every policy feasible for the joint criterion satisfies the area premise of Theorem S8.2. Since $11/80 > \delta$, a joint regret bound strictly below its threshold is impossible. Taking an infimum gives the claimed lower bound. For the upper bound, the fine-checkpoint policy has the required area bound on its simultaneous interval-coverage event, whose probability is at least $1-\delta/2$. Independently of coverage, every available arm in the family has gap at most 2ε, so $R_T \leq 2\varepsilon T$ on every sample path. The specified horizon and $J \leq M$ yield the displayed upper order, with the same universal constant as the rapid-maintenance upper bound. □

S9 Compatible-window truncated area

Proposition S9.1 (Additive truncated area in one compatible experiment)**.** *Consider finitely many batches indexed by e. Batch e contains m_e independent signs and has a deterministic validity window W_e of length H_e that begins at that batch's arrival round. During that window each sign is its arm's relative-goodness label and its boundary distance exceeds η. The signs across all batches are*

independent, other distributions do not encode them, and no information about batch e is available before its arrival. Samples remain interleaved and adaptive. Suppose every sign configuration has familywise reporting error at most $0 < \delta \le 1/16$. Let $0 < \kappa_e < \infty$ bound the two directional divergences in batch e, and define

$$h_e = \min\left\{H_e, \left\lceil \frac{m_e \log(m_e/\delta)}{1024\kappa_e} \right\rceil - 1\right\}, \qquad S = \sum_e m_e h_e.$$

When $S > 0$, one fixed full sign configuration satisfies

$$P\{\mathcal{U}_T(\eta) \ge S/8\} \ge 1/7,$$

provided every selected window lies before T and every candidate–time pair is counted only once. The windows may overlap when they concern distinct arms and their stated qualification structure is jointly realizable.

Proof. At the rth round after batch e arrives, $r \le h_e$, the entire experiment has supplied at most r samples of that batch. The contrapositive of the partial-recovery lemma shows that at least half the batch is determinate with probability less than $1/2$. Its expected unresolved count is therefore at least $m_e/4$. Let Y sum unresolved candidate–time pairs over the first h_e rounds of all selected windows. All these pairs have boundary distance larger than η, so $Y \le \mathcal{U}_T(\eta)$, while $0 \le Y \le S$ and $\overline{E}Y \ge S/4$. If $p = \overline{P}(Y \ge S/8)$, then $S/4 \le S/8 + (7S/8)p$, giving $p \ge 1/7$. Select a fixed configuration from the common product prior. Compatibility of the windows is essential: this proof cannot sum bounds from mutually incompatible reference environments. □

S10 Eventual exact classification and unconditional regret

This extension uses a separate, revisable multiresolution policy, rather than the fixed-checkpoint policy. Let epochs be $E_e = \{16^e, \dots, 16^{e+1} - 1\}$ and use cumulative targets $q_e = \lceil 4^e(e+1)^2 \rceil$. At even rounds serve, by FIFO, any arrived arm below the current target, including arms formerly reported excluded. Other rounds use full-history UCB. In epoch e report from the raw intervals with radius

$$b_{i,e}(n) = \sqrt{\frac{\log\{16i^2(e+1)^2n^2/\delta\}}{2n}},$$

clipped to $[0,1]$ and equal to $[0,1]$ at zero count. Intervals are recomputed rather than permanently intersected across epochs.

Hoeffding and a union over i, e, n bound the probability of any coverage failure by $\delta\zeta(2)^3/8 < \delta$. This proves all-time reporting reliability. Suppose arrivals eventually stop with $K < \infty$. For all large e, the current epoch's remaining checkpoint work is at most Kq_e, and its $2Kq_e$ physical service time is smaller than $15 \cdot 16^e$. Therefore each arm has at least q_{e-1} observations at the beginning of every sufficiently late epoch. Uniformly for $t \in E_e$ and actual $n \in [q_{e-1}, t]$, the radius is $O(\sqrt{\{e + \log i + \log(1/\delta)\}/q_{e-1}}) = o(1)$.

For every positive rational z, Hoeffding makes $\sum_n P(|\hat{r}_i(n) - r_i| > z) < \infty$. Tail union bounds imply only finitely many deviations of size z almost surely. Taking a countable intersection over z and the finitely many final arms gives simultaneous empirical-mean convergence. If $\min_i |\max_j r_j - r_i - \varepsilon| > 0$, then all interval comparisons are eventually correct and complete, almost surely. This conclusion is not conditioned on early coverage, and early reporting errors can be corrected.

Through $T \in E_e$, service pulls number at most Kq_e, because targets are cumulative and lifetime counts never decrease. This is $O(K\sqrt{T}(1+\log T)^2)$. Apply the unconditional UCB counting bound in Section S2 to the remaining pulls to obtain

$$ER_T = O\{K\sqrt{T}(1+\log T)^2 + \sqrt{KT\log(2T)}\} = o(T)$$

for each eventually finite instance. Exact boundary cases are not included in the complete-classification claim.

S11 General information-allocation relaxation

This result is useful beyond the separated family, but does not assert attainability. Fix a reference environment $\boldsymbol{r}$, endpoint T, and a deterministic query set

$$\mathcal{Q} \subseteq \{(i,t) : i \in \mathcal{A}_t, |\Delta_i(t) - \varepsilon| > \eta\}, \qquad W = |\mathcal{Q}|.$$

For each query choose any allowed alternative $\boldsymbol{r}'$ that reverses arm i's membership at time t. Put $u_{i,t} = P_{\boldsymbol{r}}(i \in \mathcal{V}_t)$ and

$$\Phi_\delta(u) = \begin{cases} \mathrm{kl}(1-\delta-u, \delta), & 0 \le u < 1-2\delta, \\ 0, & 1-2\delta \le u \le 1. \end{cases}$$

This function is convex and nonincreasing: the positive branch is an affine composition of binary KL, and its value and derivative vanish at the join.

Proposition S11.1 (One protocol for area and information)**.** *For any reliable policy,*

$$\sum_j E_{\boldsymbol{r}} N_j(t)\, \mathrm{kl}(r_j, r'_j) \ge \Phi_\delta(u_{i,t}).$$

If its joint area budget at $\boldsymbol{r}$ is a, then $\sum_{(i,t)\in\mathcal{Q}} u_{i,t} \le a + \delta W$. Consequently $E_{\boldsymbol{r}} R_T$ is at least the value of

$$\inf_{x,u} \sum_{s\le T} \sum_{j\in\mathcal{A}_s} \Delta_j(s) x_{s,j},$$

subject to $x_{s,j} \ge 0$, $\sum_{j\in\mathcal{A}_s} x_{s,j} = 1$, no prearrival allocations, the preceding unresolved-probability constraint, and for every query and chosen alternative,

$$\sum_{s\le t} \sum_j x_{s,j}\, \mathrm{kl}(r_j, r'_j) \ge \Phi_\delta(u_{i,t}).$$

For non-Bernoulli laws replace binary arm divergences by their distributional KL.

Proof. Let F be the event of issuing, at (i,t), the determinate label correct in the reference environment. Then $P_{\boldsymbol{r}}(F) \geq 1 - u_{i,t} - \delta$, whereas $P_{\boldsymbol{r}'}(F) \leq \delta$. The history-KL identity and data processing give the claimed constraint whenever $1 - u_{i,t} - \delta > \delta$; otherwise nonnegativity suffices. This is the precise point where the truncated KL is needed. On area success the queried unresolved sum is at most a; outside success it is at most W. Taking expectations gives $a + \delta W$. The actual occupation probabilities $x_{s,j} = P_{\boldsymbol{r}}(I_s = j)$ satisfy every constraint, with objective exactly $E_{\boldsymbol{r}} R_T$. □

One observation can enter many information constraints but is charged once in the objective. Neither convex feasibility nor an oracle allocation proves that a policy with unknown means and future arrivals attains this relaxation. Dropping δW would also change the stated risk protocol.

S12 Intermediate service configurations and disclosed-reference diagnostic

In the homogeneous repeated-update family, define

$$B_{J,M} = 2(M+1)\{J(2M+1) + J(J-1)/2\} + 2(M+J+1).$$

If a predetermined set of s of the J candidate batches uses precision $\rho_f = d/4$ and the rest uses $\rho_c = v/8$, the shared-evidence bound gives

$$\mathcal{U}_T(\eta) \leq (J-s)MH + B_{J,M} q_f$$

on checkpoint coverage. The unconditional reward bound gives

$$ER_T \leq 2\varepsilon M\{sq_f + (J-s)q_c\} + 64\left(\frac{JM}{\varepsilon} + \frac{J^2}{v}\right)\log(2T) + 1.$$

Every adjacent precision pair satisfies $d + \rho_e + \rho_{e+1} < v$, and every window completes its checkpoints, so the proof follows by applying the same batchwise accounting. The choice of which batches receive fine service must be made without their reward data. These are attainable points, not a proof of optimality at every area budget.

An additional diagnostic gives each batch's reference identity to the policy and discloses that its role is best within its own window. Its numerical mean remains unknown. Sample the reference to its checkpoint, then its candidates to theirs, and otherwise sample the latest reference. Keep the same reporting comparisons. For windows long enough to complete this work, the regret in the specified family is pathwise

$$R_T = \sum_{e,k} (\varepsilon - \sigma_{e,k} d) q_{e,k}.$$

Reference sampling and idle exploitation have zero regret, whereas each candidate checkpoint occurs in its birth window and pays its displayed gap. The coarse version has $R_T \leq 2\varepsilon JMq_c$ with no

extra $\log T$ reward term. Comparing with the rapid lower bound isolates a squared v/d cost ratio. This additional role information makes the allocation problem easier; this policy is not the main unknown-reference algorithm. Its interval-based label reliability still holds on the full bounded model, while its reward statement only holds on the disclosed-role performance family.

S13 Computational details

S13.1 Configuration, records, and implementation names

The numerical study has four environments, nine methods, and 160 paired replications per environment, for 5,760 runs of 20,000 physical pulls. The main table displays six methods; the cost–area figure and Table 1 include all nine. The additional path figure below uses the same six methods as the main path figure.

The local reproducibility materials contain the complete configuration, run-level records, saved paths, summary statistics, and paired differences in `experiments/results/`. The configuration gives every arrival time and mean, each method's parameters, and the seed mechanism. For replication seed s in environment index j, the generator is NumPy PCG64 initialized with `SeedSequence([s,j])`, where $s = 202610150, \ldots, 202610309$ and $j = 0, \ldots, 3$. The resulting Bernoulli potential streams are shared across methods. The policy uses only observed rewards and current arrivals. Array preallocation is simulation storage and does not expose future arms or the evaluation endpoint to a policy.

The reporting configuration is named Checkpoint-UCB in the article. For reproducibility, its unchanged reference interface is `CheckpointUCBAA` in `reference_cp_ucb_aa.py`. The experiment driver is `run_experiments.py`. The data IDs `CP-FIFO-fine`, `CP-FIFO`, `CP-confidence`, and `CP-period8` retain their existing meaning. In particular, the first two are FIFO service every second pull with $\rho = .04$ and .12; the latter two change allocation as specified in the main article. The remaining IDs are `UCB-checkpoint`, `UCB-prefix`, `Growing-MOSS`, `ALL-epsilon-adapt`, and `Batch100-UCB`. The ALL-ε rule is a specified retained-history adaptation, not a reproduction of FAREAST or $(\mathrm{ST})^2$.

`runs.csv` records one row per environment, seed, and method, including full-run label errors, both unresolved-area totals, service counts and regret, final per-arm counts, and the regret components below. `paths.npz` contains all nine methods' regret, unresolved counts, cumulative area, and total unresolved counts on a common grid, with every arrival included. This is a saved time grid, not a per-round action log. `summary.csv` and `paired_differences.csv` provide the numerical summaries; the two table source files are generated from these records. `figures/gen_fig_performance.py` is the plotting and tabulation entry point.

S13.2 Admission and reward-retirement accounting

Let $\mathcal{A}_t^{\mathrm{adm}} \subseteq \mathcal{A}_t$ be the arms admitted for reward selection, and let $B_t \subseteq \mathcal{A}_t^{\mathrm{adm}}$ be its nonempty reward-active subset. Assume $I_t \in B_t$. Adding and subtracting the admitted and active best means

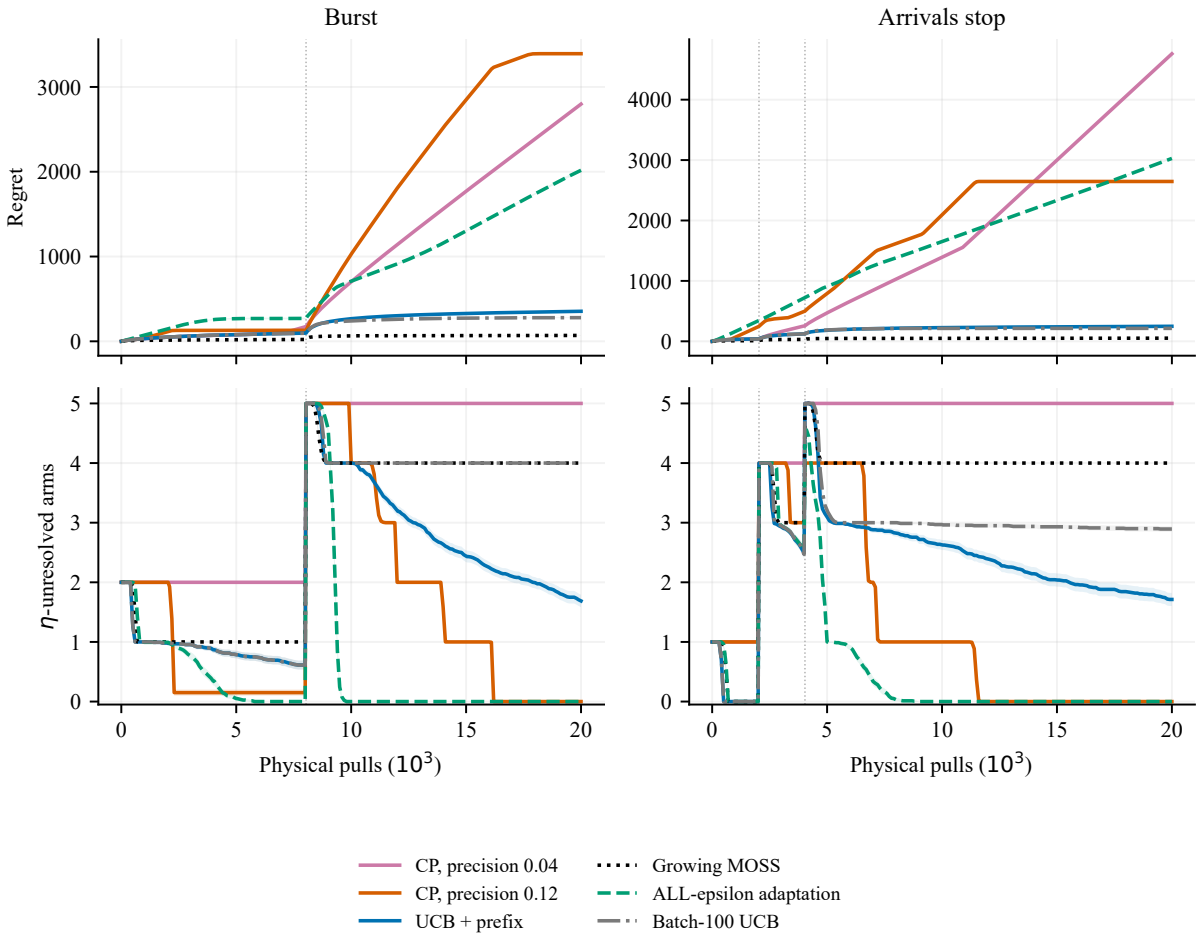


Figure 1: Additional paths for the burst and stopped-arrival environments. The six displayed policies, common potential reward streams, 160 replications, and 1.96 Monte Carlo standard error bands are the same as in the main path figure. Vertical lines mark actual arrivals. The full run-level data and saved paths retain all nine policies.

Table 1: Allocation diagnostics. Each cell gives mean forced service pulls and mean final η-unresolved arms. UCB and MOSS have zero forced service.

Method	Stable	Improvements	Burst	Stop
CP-FIFO ($\rho = .04$)	10000, 8.00	10000, 6.00	10000, 5.00	10000, 5.00
CP-FIFO ($\rho = .12$)	7340, 0.16	5786, 0.00	5991, 0.00	5737, 0.00
CP-confidence	7144, 0.16	4653, 0.00	6092, 0.00	5322, 0.00
CP-period8	2500, 6.74	2500, 0.20	2225, 3.00	2500, 2.00
UCB-checkpoint	0, 8.00	0, 3.00	0, 4.00	0, 4.00
UCB-prefix	0, 6.23	0, 0.11	0, 1.69	0, 1.71
Growing-MOSS	0, 7.00	0, 1.00	0, 4.00	0, 4.00
ALL-ε adapt.	3696, 0.00	10000, 0.00	7794, 0.00	10000, 0.00
Batch100-UCB	0, 6.40	0, 0.53	0, 4.00	0, 2.89

at each physical pull gives the pathwise identity

$$\begin{aligned} R_T = &\sum_{t\le T}\big(\max_{i\in B_t} r_i - r_{I_t}\big) \\ &+ \sum_{t\le T}\big(r^*(t) - \max_{i\in\mathcal{A}_t^{\mathrm{adm}}} r_i\big) + \sum_{t\le T}\big(\max_{i\in\mathcal{A}_t^{\mathrm{adm}}} r_i - \max_{i\in B_t} r_i\big). \end{aligned} \tag{4}$$

The components are within-active learning, admission delay, and the historical-retirement deficit. They are all nonnegative. In Batch100-UCB, admission occurs at pulls $1, 101, 201, \ldots$, before action selection. Reward retirement uses the confidence comparison among currently admitted active arms after the current observation. Reporting continues to cover all physically arrived arms, with $[0, 1]$ intervals before admission. Neither waiting for admission nor reward retirement is itself a certificate of membership or nonmembership in the near-optimal set.

For Checkpoint-UCB, every arrived arm is immediately reward-eligible, and negative reporting does not remove it from reward selection. Thus $B_t = \mathcal{A}_t^{\mathrm{adm}} = \mathcal{A}_t$, so the last two components vanish identically. The same statement applies to the other immediate-admission methods that retain every arm for reward selection.

In the Batch100-UCB records, the admission contributions are 0, 33.92, 16.00, and 22.40 for the stable, improving, burst, and stopped-arrival environments, respectively. No retirement deficit occurred. Across all 5,760 records, the largest absolute difference between regret and the sum in (4) is below 3.1×10^{-12}.

S13.3 Verification and computational resources

The production manifest records Python 3.11.7, NumPy 1.26.4, Numba 0.59.0, Windows 10, and 13.5 seconds for production generation and recording, excluding initial compilation and subsequent plotting. The resource snapshot records 20 physical CPU cores, 28 logical cores, and 31.71 GB system memory. The small physical-round loops use compiled CPU execution. A detected GPU was not needed. Dependencies and hashes of the reference implementation, driver, configuration, and numerical records accompany the materials.

The saved production parity check compared the compiled FIFO procedure with the reference

interface for 20,000 consecutive rounds, checking actions, service indicators, available-arm counts, and reports. That is a historical implementation check. In this revision, separate short deterministic checks compare the same interfaces around arrivals, quota crossings, frozen intervals, and negative reports. The revision also recomputes the 24 main-table rows, the complete allocation table, and the 640 all-unresolved finer-checkpoint records from the saved data. These operations leave the simulation records and policy code unchanged. They are implementation and accounting checks, rather than new method comparisons or probability certificates.

The local materials support these resource and reproducibility descriptions. A public repository or permanent release identifier has not been verified for this version.

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